Universidade de São Paulo
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Support Vector Machines

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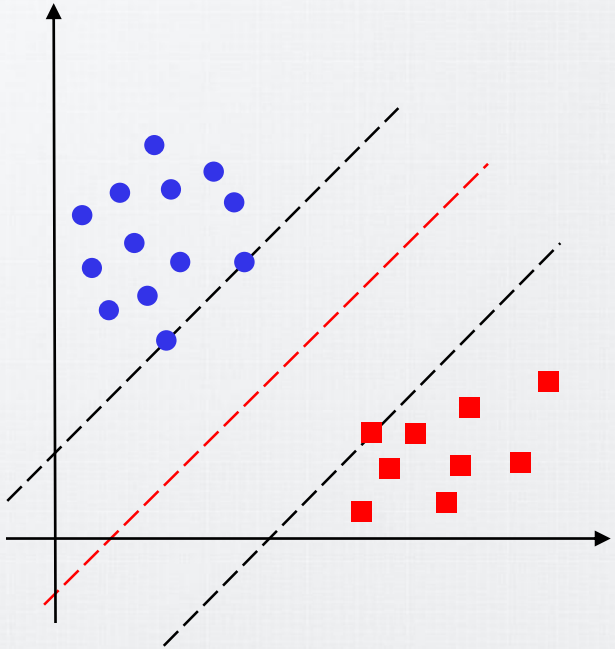
Introduction

Support Vector Machines

- The Support Vector Machine (SVM) model class was the most popular approach for *off-the-shelf* supervised learning
  - When we don't have any specialized prior knowledge about a domain

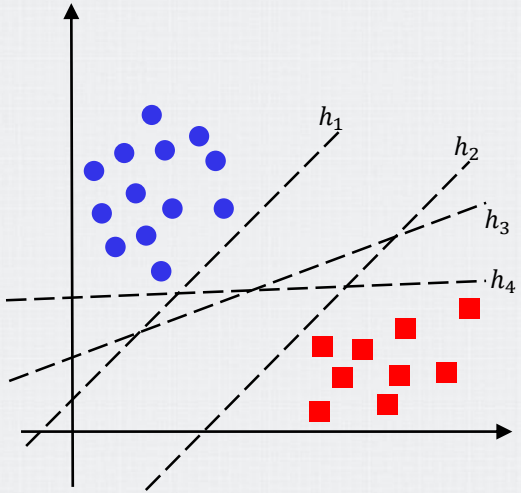


Intuition

Support Vector Machines

- Given that there are many possible linear classifiers (hyperplanes), which of the infinite hyperplanes should we choose?

- The key insight of SVMs is that some training examples are more informative than others, and the algorithm exploits this fact to determine the separating hyperplane
  - Paying attention to them can lead to better generalization



Properties

Support Vector Machines

- SVMs construct a maximum margin separator

- SVMs create a linear separating hyperplane
  - They have the ability to embed the data into a higher-dimensional space, using the so-called **kernel trick**

- SVMs (non primal) are nonparametric
  - The separating hyperplane is **defined** by a set of example points, not by a collection of parameter values
  - SVM model **keeps only the samples** that are closest to the separating plane

Preliminaries

Support Vector Machines

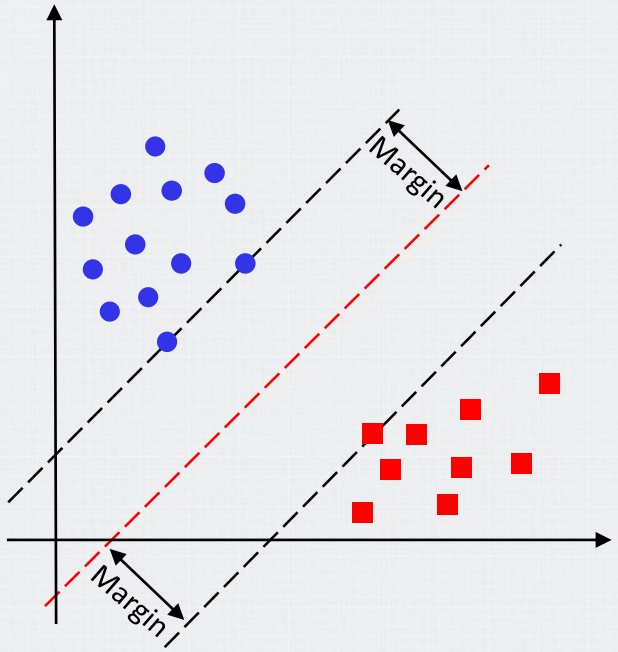
- Let $w \in \mathbb{R}^m$ be a hyperplane and b a real number
  - While we could put the intercept into the weight vector w, SVMs keep the intercept as a separate parameter, b

- Given w and b, the SVM estimates the label in terms of $y = sign\ (w \cdot x + b)$
  - (Canonical) SVMs use the convention that class labels are $+1$ and -1
  - $y \in \{+1, -1\}$

Preliminaries

Support Vector Machines

- The margin is the distance between the hyperplane and the closest training samples on either side
  - A large margin contributes to a better generalization
  - Therefore, we want to **maximize** the margin



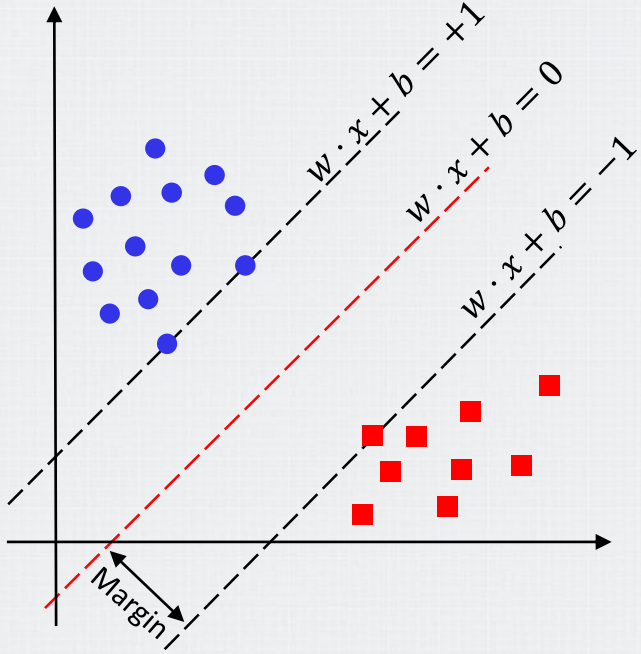
Preliminaries

Support Vector Machines

- The separating hyperplane consists of all points x that satisfy $\{x : w \cdot x + b = 0\}$
  - Decision boundary

- $w \cdot x + b = +1$
  - Margin of the positive class $(+1)$

- $w \cdot x + b = -1$
  - Margin of the negative class (-1)

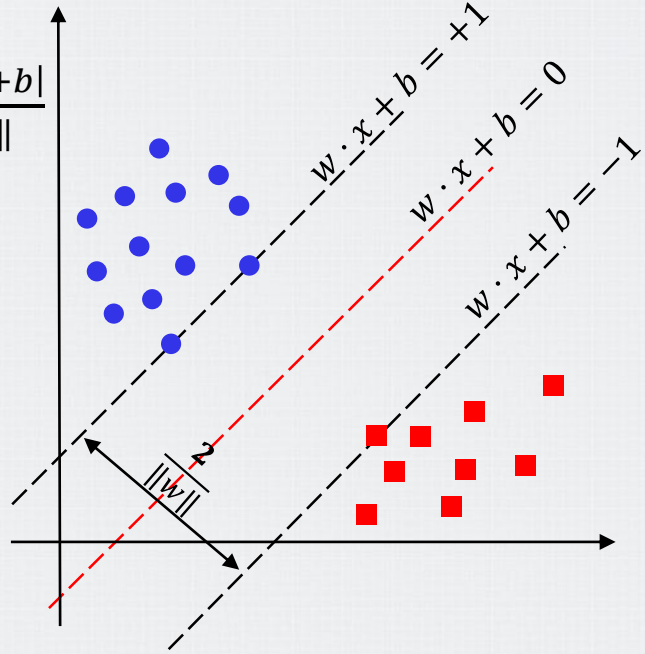


Preliminaries

Support Vector Machines

- Geometrically speaking, the equations $w \cdot x + b = +1$ and $w \cdot x + b = -1$ define two parallel hyperplanes
  - **Support vectors lie on the margin hyperplanes**

- Given that the distance from a sample to w is $\frac{|w \cdot x + b|}{\|w\|}$
  - When $w \cdot x + b = +1 \Rightarrow \frac{|1|}{\|w\|} = \frac{1}{\|w\|}$
  - When $w \cdot x + b = -1 \Rightarrow \frac{|-1|}{\|w\|} = \frac{1}{\|w\|}$
  - **Therefore, the distance between these hyperplanes is given by $\frac{2}{\|w\|}$**



Preliminaries

Support Vector Machines

- $w \cdot x + b > 0$
  - x belongs to class $+1$
- $w \cdot x + b < 0$
  - x belongs to class -1
- $\frac{|w \cdot x + b|}{||w||}$
  - Distance from the hyperplane

Hands-on

Problem Definition

Support Vector Machines

- The distance between the hyperplanes is $\frac{1}{\|w\|}$, so the smaller the norm $\|w\|$, the **larger the distance between them**
  - Remember that the SVM seeks the separating hyperplane with **maximum margin**

- We can rewritte $w \cdot x + b \geq 1$ for $y = +1$ and $w \cdot x + b \leq -1$ for $y = -1$ as
  - $y(w \cdot x + b) \geq +1$
  - Note that $(w \cdot x + b) \leq -1 \Rightarrow -(\mathrm{w} \cdot x + b) \geq +1$

Problem Definition

Support Vector Machines

- Given the previous definitions, the SVMs constraints are naturally
  - $max\ \frac{1}{\|w\|}$ subject to $y_i(w \cdot x_i + b) \geq +1, \forall x_i \in X$
  - Since maximizing $\frac{1}{\|w\|}$ is equivalent to minimizing $\frac{1}{2}\|w\|^2$, in the separable case, SVM solves the following convex optimization problem

$$\boldsymbol{min\ \frac{1}{2}\|w\|^2 \text{ subject to } y_i(w \cdot x_i + b) \geq 1, \forall x_i \in X}$$

Quadratic Programming

Support Vector Machines

- $min\frac{1}{2}\|w\|^2$ subject to $y_i(w \cdot x_i + b) \geq 1, \forall x_i \in X$ is an instance of standard quadratic optimization problem
  - The complexity depends on feature dimension

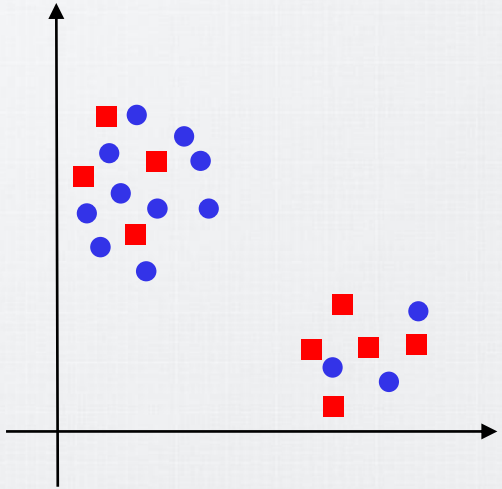
- A variety of commercial and open-source solvers are available for solving convex QP problems

- Lagrange multipliers (α)
  - $L = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i(y_i(w \cdot x_i + b) - 1)$
  - We minimize w.r.t. primal variables $\boldsymbol{w}$ and $\boldsymbol{b}$ (**SVM primal**)
  - The saddle point gives the solution
  - Saddle point conditions: $\frac{\partial}{\partial w}L(w, b, \alpha) = 0, \frac{\partial}{\partial b}L(w, b, \alpha) = 0$

Non-Separable Case

Support Vector Machines

- If the data is not linearly separable, the previous algorithm will not work
  - If the two classes **are not linearly separable** such that there is **no** hyperplane to separate them
  - The two classes are not separable by a hyperplane
  - In most practical settings, the training data is not linearly separable

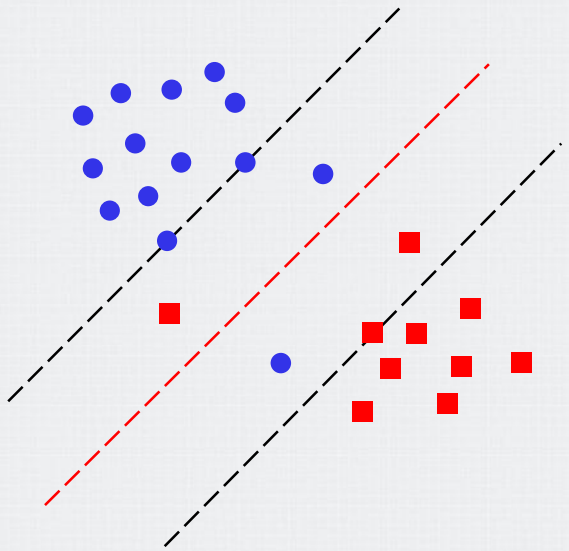


Soft Margin Hyperplane

Support Vector Machines

- When the data **is not linearly separable,** we instead allow some samples to be on the incorrect side of the region of maximum margin
  - Or even the incorrect side of the hyperplane

- The margin is soft because some of the training samples can violate it

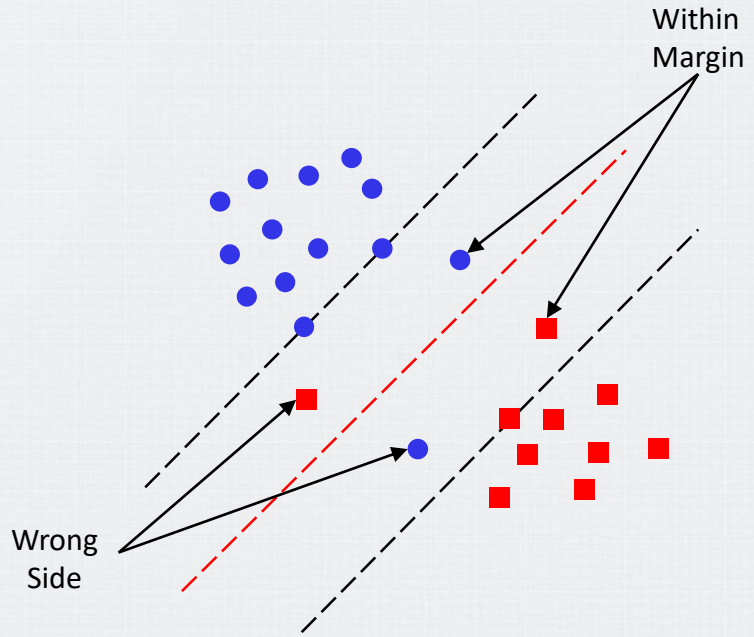


Soft Margin Hyperplane

Support Vector Machines

- For the non-separable case, we introduce slack variables ξ_i to relax the constraints of the canonical hyperplane equation

- There are two types of deviation
  - A sample lies on the wrong side of the hyperplane (misclassified)
  - A sample lies on the correct side but within the margin



Soft Margin Hyperplane

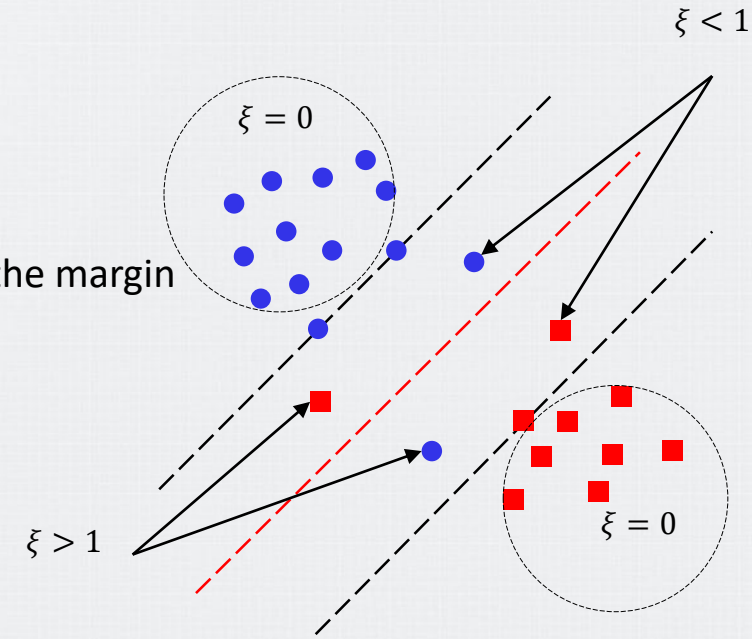
Support Vector Machines

- To address the previous problem, we relax the original optimization problem to
  - $y_i(w \cdot x_i + b) \geq 1 - \xi_i$

- $\xi_i = 0$
  - No problem (its is the original optimization)

- $0 < \xi_i \leq 1$
  - The model correctly classifies x_i but inside the margin

- $\xi_i > 1$
  - The model misclassifies x_i



Soft Margin Hyperplane

Support Vector Machines

- The soft error becomes $\sum_i \xi_i$
- Adding $\sum_i \xi_i$ to the SVM constraints, we obtain

$$min\,\frac{1}{2}\|w\|^2 + \boldsymbol{C}\sum_{\boldsymbol{i}}\boldsymbol{\xi_i} \text{ subject to } y_i(w \cdot x_i + b) \geq 1 - \boldsymbol{\xi_i}, \forall x_i \in X$$

  - This is the standard SVM of popular machine learning libraries
  - $\left|\{\xi^i > 1\}\right|$
    - An upper bound of the number of misclassifications
  - $\left|\{\xi^i > 0\}\right|$
    - An upper bound of the number of number of margin violations

Soft Margin Hyperplane

Support Vector Machines

- C is the penalty factor
  - A hyperparameter we tune

- When C is small, the model allows **more margin violations**
  - The margin tends to be **wider**

- When C is larger, the model **penalizes violations more heavily**
  - The margin tends to be narrower

- In the limit $C \to \infty$, we recover the earlier

SVM for separable data

Hands-on

Final Notes

Support Vector Machines

- After estimating w and b, the prediction for a given sample x_i
  - $sign(w \cdot x_i + b)$

- Therefore, what is the role of support vectors ($\{x : w \cdot x + b \pm 1\}$) during prediction?

Kernel Trick

Motivation

Kernel Trick

- When the problem is nonlinear, instead of trying to fit a nonlinear model, we can map the problem to a **new space** (**nonlinear** transformation – $\phi(.)$) and then use a **linear model** in this new space
  - The linear model in the new space corresponds to a nonlinear model in the original space

Hands-on

Problem Definition

Kernel Trick

- The problem is the same as in the linear SVM, except that now we define the constraints in the new feature space ($\phi(.)$)

$$min\,\frac{1}{2}\|w\|^2 + C\sum_i \xi_i \text{ subject to } y_i(w \cdot \boldsymbol{\phi}(\boldsymbol{x_i}) + b) \geq 1 - \xi_i, \forall x_i \in X$$

- Kernelization technique (Kernel trick)
  - We never need to compute $\phi(x_i)$ explicitly
  - Since SVM only depend on inner products, we replace $\phi(x_i) \cdot \phi(x_i')$ with a kernel function $K(x_i, x_i')$

Definitions

Kernel Trick

- A kernel is a function that quantifies the similarity of two observations
  - $K(x_i, x_j) = \sum_{z=1}^{m} x_i^z x_j^z$
  - z stands for the zth feature

- $\boldsymbol{K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle}, \forall x_i, x_j \in X$
  - $\langle \phi(x_i), \phi(x_j) \rangle \Rightarrow \phi(x_i) \cdot \phi(x_j)$

Example

Kernel Trick

- Let $\phi(x) = \begin{bmatrix} x^1x^1 \\ x^1x^2 \\ x^2x^1 \\ x^2x^2 \end{bmatrix}$

- Let $K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j) = \begin{bmatrix} x_i^1x_i^1 \\ x_i^1x_i^2 \\ x_i^2x_i^1 \\ x_i^2x_i^2 \end{bmatrix} \cdot \begin{bmatrix} x_j^1x_j^1 \\ x_j^1x_j^2 \\ x_j^2x_j^1 \\ x_j^2x_j^2 \end{bmatrix} = \cdots = \boldsymbol{(x_i \cdot x_j)^2}$

For the sake of simplicity

- Observe that instead of mapping to 4 dimensions ($\phi(.)$) and computing the inner product, we simply compute $\boldsymbol{(x_i \cdot x_j)^2}$

Inference

Kernel Trick

- Assume a single sample x and S the set of support vectors ($S \subset X$)

- Linear
  - $w = \sum_{i=1}^{S} \alpha_i y_i x_i$
  - $f(x) = w \cdot x + b$

- Kernel
  - $w = \sum_{i=}^{S} \alpha_i y_i K(x_i, x)$
  - $f(x) = w + b$

Hands-on

Practical Issues

Kernel Trick

- Algorithms that only depend on inner products between sample points

- Replacing the original inner product in the input space with positive definite kernels immediately extends algorithms such as SVMs to a linear separation in that high-dimensional space
  - Equivalently, to a non-linear separation in the input space

Popular Kernel

Kernel Trick

- Polynomial
  - $K(x_i, x_j) = \left(1 + \sum_{k=1}^{m} x_i^k x_j^k\right)^d$
  - d is positive integer – hyperparameter

- Radial
  - $K(x_i, x_j) = \exp(-\gamma \sum_{k=1}^{m}((x_i^k - x_j^k)^2)$
  - γ is a positive constant – hyperparameter

Hands-on

Multi-class Algorithms

Motivation

Multi-class Algorithms

- In order to deal with multi-class tasks, some approaches reduce the problem to that of multiple binary classification tasks

- One-Versus-All

- One-Versus-One

One-Versus-All

Multi-class Algorithms

- This technique consists of learning k **binary classifiers**
  - One-versus-the-rest technique

- The technique seeks to discriminate one class from all the others
  - For this purpose, we need to relabel the dataset C times
  - Each time, the label from class c is 1 and the rest is -1
  - We train a binary classifier $f_c(.)$ for each combination

- Prediction
  - $\hat{y} = argmax_{c \in C}\, f_c(x)$

Hands-on

One-Versus-One

Multi-class Algorithms

- This technique consists of using the training data to learn (independently) a binary classifier for each pair of distinct classes
  - Therefore, it requires training $\binom{k}{2} = k(k-1)/2$ binary classifiers

- Prediction
  - Majority voting
  - $\hat{y} = argmax_{c\ in\ C} \sum_{i=1}^{k} I(\hat{y_i} = c)$
  - I is the indicator function

$f_1(.)$	Class 1 vs. Class 2
$f_2(.)$	Class 1 vs. Class 3
$f_3(.)$	Class 2 vs. Class 3

Hands-on

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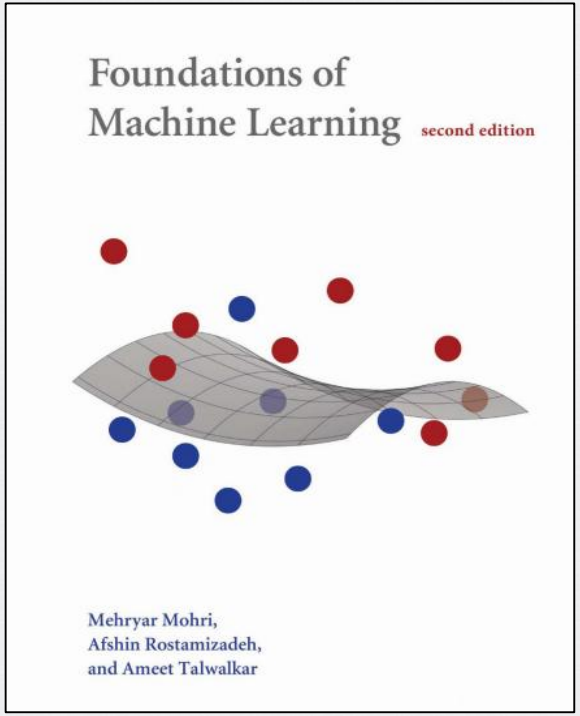
    - 9.4 Aggregated multi-class algorithms



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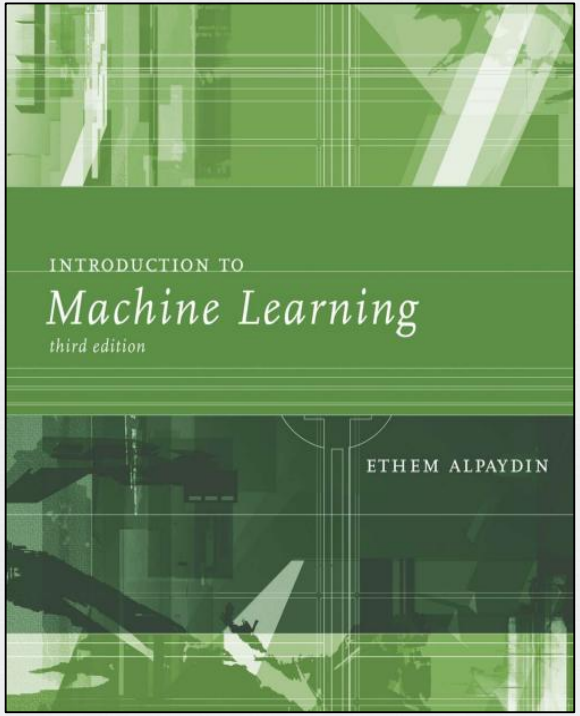

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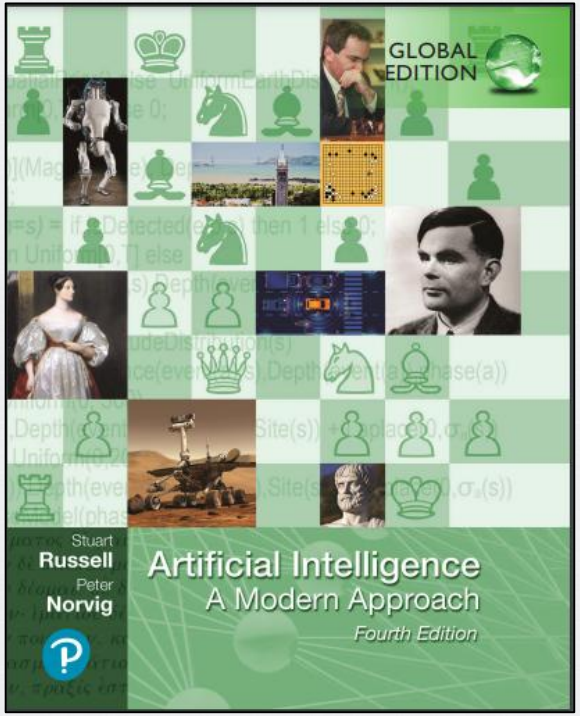

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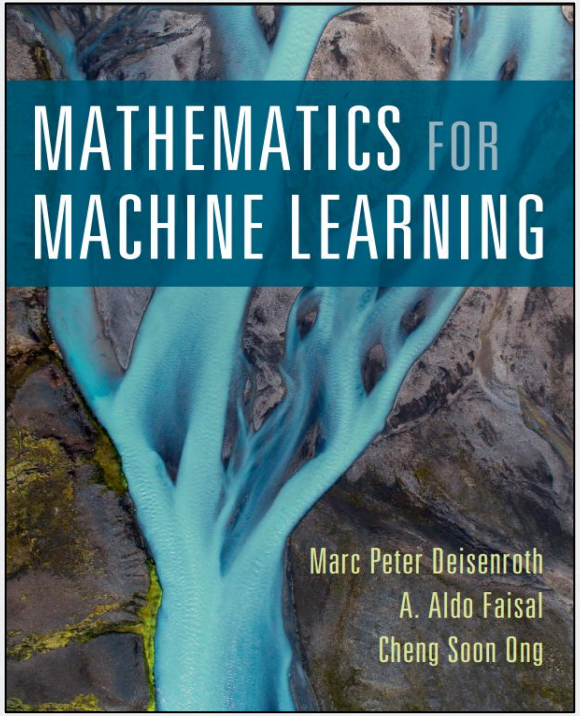
  - Chapter 7
    - 7.2 Constrained Optimization and Lagrange Multipliers



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